

```

R<x, y, z, w> := PolynomialRing;

C := Curve(ProjectiveSpace(Rationals(), 3), [f1, f2]);

Genus(C);

pt := C![0, -1, 1, 0];

IsSingular(pt);

// yields E along with phi, a map from C to E
E, phi := EllipticCurve(C, pt);

chk, phi2 := IsInvertible(phi);

G, mp := MordellWeilGroup(E);

// get coordinates of first generator in G
mp(G.1)

// single variable ring for finding roots
R<x> := PolynomialRing(Rationals());

Roots(f);

for d in Divisors(176) do;
    ...
end for;

N^2 = dM^4 + aM^2e^2 + (b/d)e^4

IsSquarefree(d);

[ ... ] cat [ ... ];

f := _x^4 - _;
C := HyperellipticCurve(f);

IsLocallySolvable(GenusOneModel(C));

// factor polynomial f
Factorization(f);

Points(C : Bound := 10000);

```

```
PointSearch(C, 100);
```

```
Rank(C);
```

```
RootNumber(C);
```

```
L := LSeries(C : Precision := 5);
```

```
Evaluate(L, 1 : Derivative := 1)
```